SeedingMinds

Assessment Name

Java_S1_271225

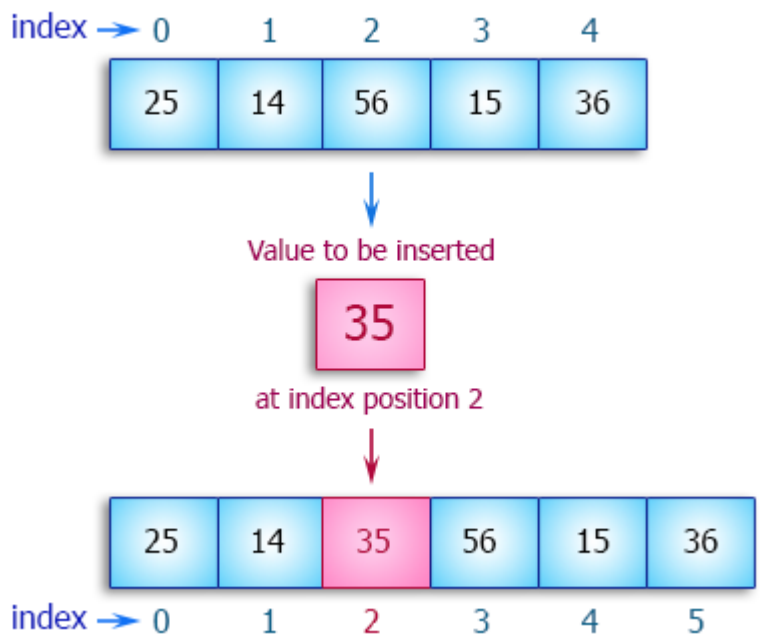
Module: Coding

Question 1:

Marks: +10 -0

insert an element (specific position) into an array

Write a Java program to insert an element (specific position) into an array.



Constraints:

N/A

Example:

Input:

25, 14, 56, 15, 36, 56, 77, 18, 29, 49

Output:

25, 14, 5, 56, 15, 36, 56, 77, 18, 29

Explanation:

-

Public Test Cases:

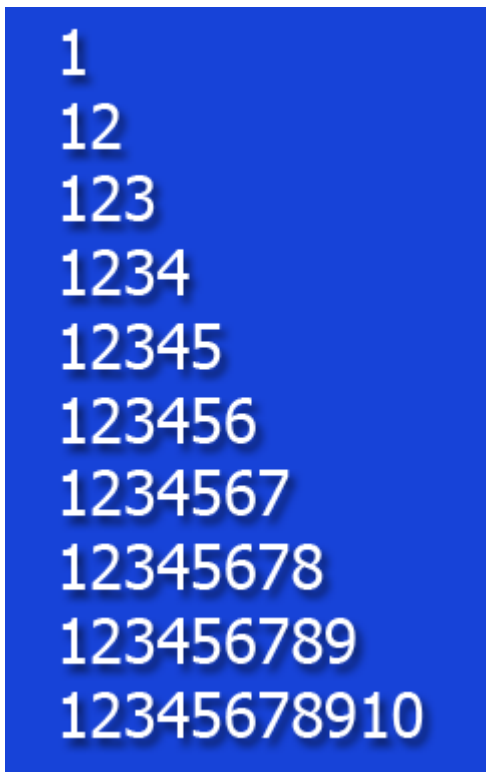
#	INPUT	EXPECTED OUTPUT
1	25, 14, 56, 15, 36, 56, 77, 18, 29, 49	25, 14, 5, 56, 15, 36, 56, 77, 18, 29

Question 2:

Marks: +10 -0

pattern like right angle triangle with a number.

Write a program in Java to display the pattern like right angle triangle with a number.



Constraints:

N/A

Example:

Input:

```
10

Output:
1                                                                    1
12
123
1234
12345
123456
1234567
12345678
123456789
12345678910
```

Explanation:

-

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	10	1 12 123 1234 12345 123456 1234567 12345678 123456789 12345678910

Question 3:

Marks: +10 -0

Password Validation

Write a program to check/validate given password is perfect or not ?

Constraints:

Input must be a string and it should contain **9 characters** or else Print **Not a Perfect Password.**

Atleast one character should be **small** or else Print **Not a Perfect Password.**

Atleast one character should be **capital** or else Print **Not a Perfect Password.**

Password should contain atleast **one Number** or else Print **Not a Perfect Password.**

Password should contain atleast **one Special Character**(Except Space) or else Print **Not a Perfect Password.**

Example:

Input 1 : abCde@123

Output 1 : Perfect Password

Input 2 : 0123456789

Output 2 : Not a Perfect Password

Explanation:

NA

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	abCde@123	Perfect Password
2	0123456789	Not a Perfect Password

Question 4:

Marks: +10 -0

Steel Strength Validation

A certain grade of steel is graded according to the following conditions:

i. Hardness must be greater than 50

ii. Carbon content must be less than 0.7

iii. Tensile strength must be greater than 5600

The grades are as follows:

- Grade is 10 if all three conditions are met
- Grade is 9 if conditions (i) and (ii) are met
- Grade is 8 if conditions (ii) and (iii) are met
- Grade is 7 if conditions (i) and (iii) are met
- Grade is 6 if only one condition is met
- Grade is 5 if none of the conditions are met

Write a program, which will require the user to give values of hardness, carbon content and tensile strength of the steel under consideration and output the grade of the steel.

Constraints:

Input Format:

Line1: 3 floating point values separated by space (Hardness, Carbon content, Tensil strength respectively)

Output Format:

Line 1: Integer number (represents the grade)

Example:

Input:

55 0.5 4500

Output:

9

Explanation:

Explanation:

Hardness = 55

Carbon content = 0.5

Tensile strength = 4500

According to the inputs given, only condition 1 and condition 2 are satisfied hence the grade is 9

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	55 0.5 4500	9
2	55 0.5 8000	10